ROHAN PATRICK JR

Monticello, New York • rpatrick@albany.edu • https://github.com/rohanpatrick568 • rohanpatrick568.github.io

EDUCATION

Bachelor of Science in Electrical & Computer Engineering *State University of New York at Albany* – GPA: 3.1/4.0
Expected **Dec 2026**

Minor in Mathematics & Computer Science

- *Dean's List of Distinguished Students*
- *National Society of Black Engineers*
- *UAlbany Marching & Spirit Band*
- *Awarded 3rd Place in the 2025 IEEE Hack-A-Damien Hackathon*

SKILLS

- **Technical:** Python, Systems Programming (C/C++), Java, Assembly (MIPS), VHDL, Embedded C, ISA Design, Digital Signal Processing (DSP) Fundamentals, Network Protocols (TCP/IP)
  Tools: Xilinx Vivado (VHDL), MATLAB, PyTorch, PSPICE, AutoCAD/Revit, Git/GitHub, FreeCAD, Shapr3D
- **Course Work: Hardware & Systems:** Computer Architecture & Organization (A), Digital Systems (VHDL), Programming Hardware Systems (TI MSP430), Signals and Systems (B+), Advanced Electronics. **Advanced Computing:** Deep Learning (B), Data Structures (B+), Computer Communication Networks (B), Systems Analysis and Design (B+). **Engineering Mathematics:** Probability & Statistics for Engineering (A), Linear Algebra, Ordinary Differential Equations, Electrical Energy Systems (Spring 2026).

SELECTED PROJECTS

Haptics-Based Assistive Navigation System **June – Aug 2025**
Research Developer | Raspberry Pi

- Developed a prototype assistive device that uses haptic signals to guide visually impaired users in navigation tasks.
- Programmed microcontrollers to interface with sensors and vibration motors, creating real-time tactile feedback.
- Improved accessibility and independence by offering a non-visual alternative to traditional guidance tools.
- Link to article: `https://www.albany.edu/news-center/news/2025-undergrad-research-project-develops-haptics-assist-visually-impaired`

Electronic Memory Game Project **Sept – Nov 2024**
Solo Developer | C (Programming Hardware Systems)

- Developed an embedded game system (like Mario) in C on an MSP430 microcontroller through UART protocols.
- Utilized buttons and a thumbstick for user interactions, including random sequence generation and user input validation.
- Developed and programmed game states, including initialization, random sequence generation, user input validation, error handling, and victory detection; ensured seamless gameplay experience across 5 distinct phases on MSP430 microcontroller.

Deep Learning Image Classification Project **July – Aug 2024**
Solo Developer | C

- Created Deep Learning Algorithm using custom C code to classify handwritten digits (0-9) from the MNIST dataset containing 60,000 images for training.
- The model is a neural network totaling 858 neurons which process the input images to predict the corresponding digit. Implemented the gradient descent function to adjust the neurons during training to improve the network's accuracy.
- The results after testing the trained model on 10,000 new images showed an accuracy of >98%.

EXPERIENCE

SUNY Research Foundation, *Albany, NY* **June – Aug 2025**
Undergraduate Researcher

- Designed and developed a wearable haptic feedback system to assist visually impaired individuals with navigation.
- Integrated embedded systems and haptic actuators to translate visual and spatial data into tactile cues.
- Conducted extensive user testing and iterative prototyping, collaborating with faculty to refine usability.
- Research featured in the UAlbany News Center (2025) as part of the university's Undergraduate Research program.

SUNY at Albany Residential Life, *Albany, NY* **Aug 2023 – Present**
Resident Assistant

- Upholds a professional, safe, and fun environment for first year students at the University at Albany. Serving as a leader and authority for 40 residents in addition to providing support for other RAs as needed.
- Provided customer service and executed various administrative tasks while managing office operations.
- Conducted interviews for the hiring process, providing feedback and recommendations to supervisors to support final hiring decisions.

Acuta Digital, Inc., *New York, NY* **July – Aug 2023**
Technology Development Intern

- Supported the IT team with developing and maintaining software and systems for a full-service digital marketing and software development firm.
- Assisted with data entry and data migration tasks, and provided online research support for the market and sales team.
- Collaborated with team leadership to accomplish client tasks and provided administrative support, including taking meeting notes.
- Conducted remote research independently while remaining deadline-driven across all assigned projects.